

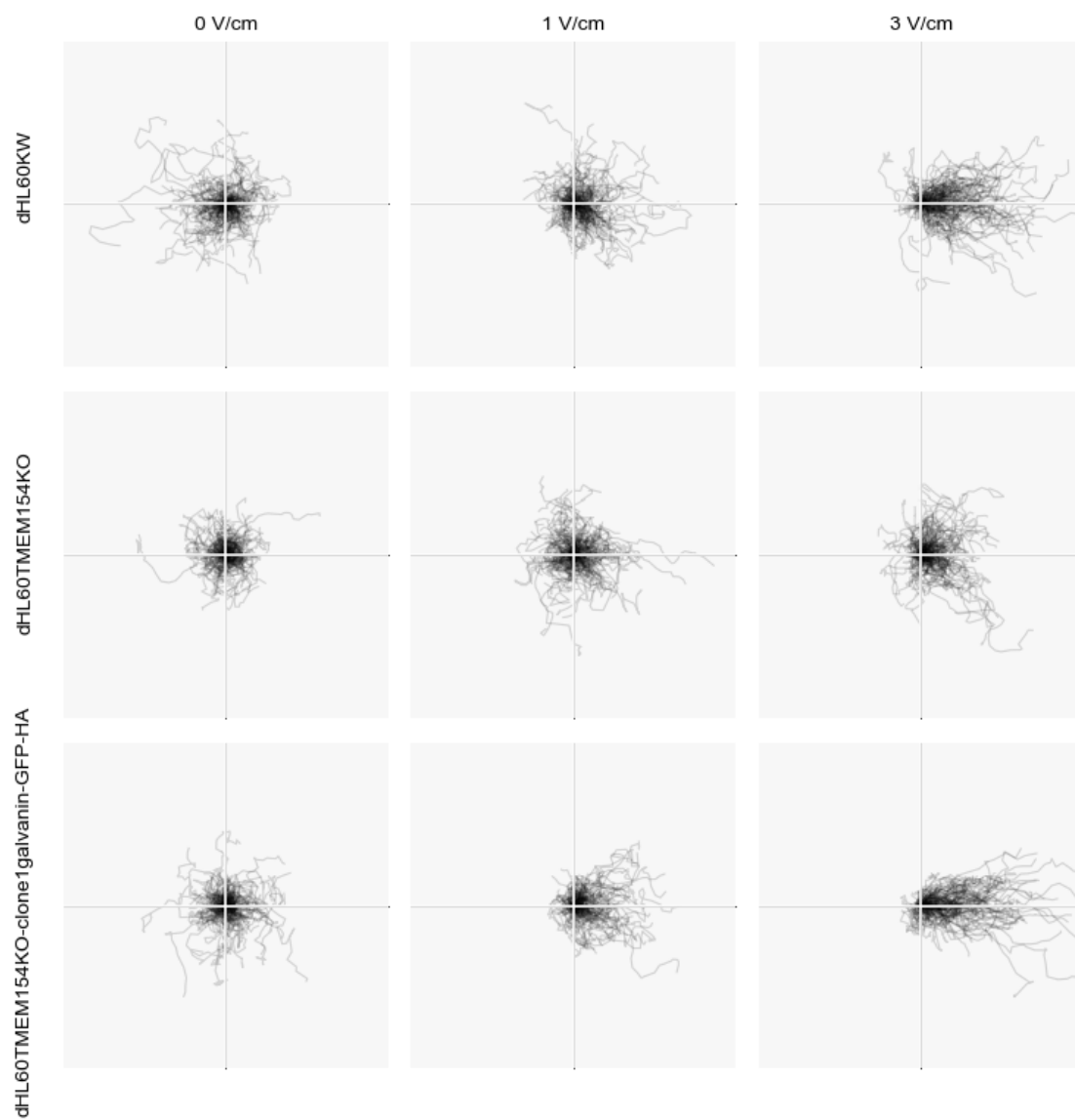
2025__HL60__KO__rescue__data

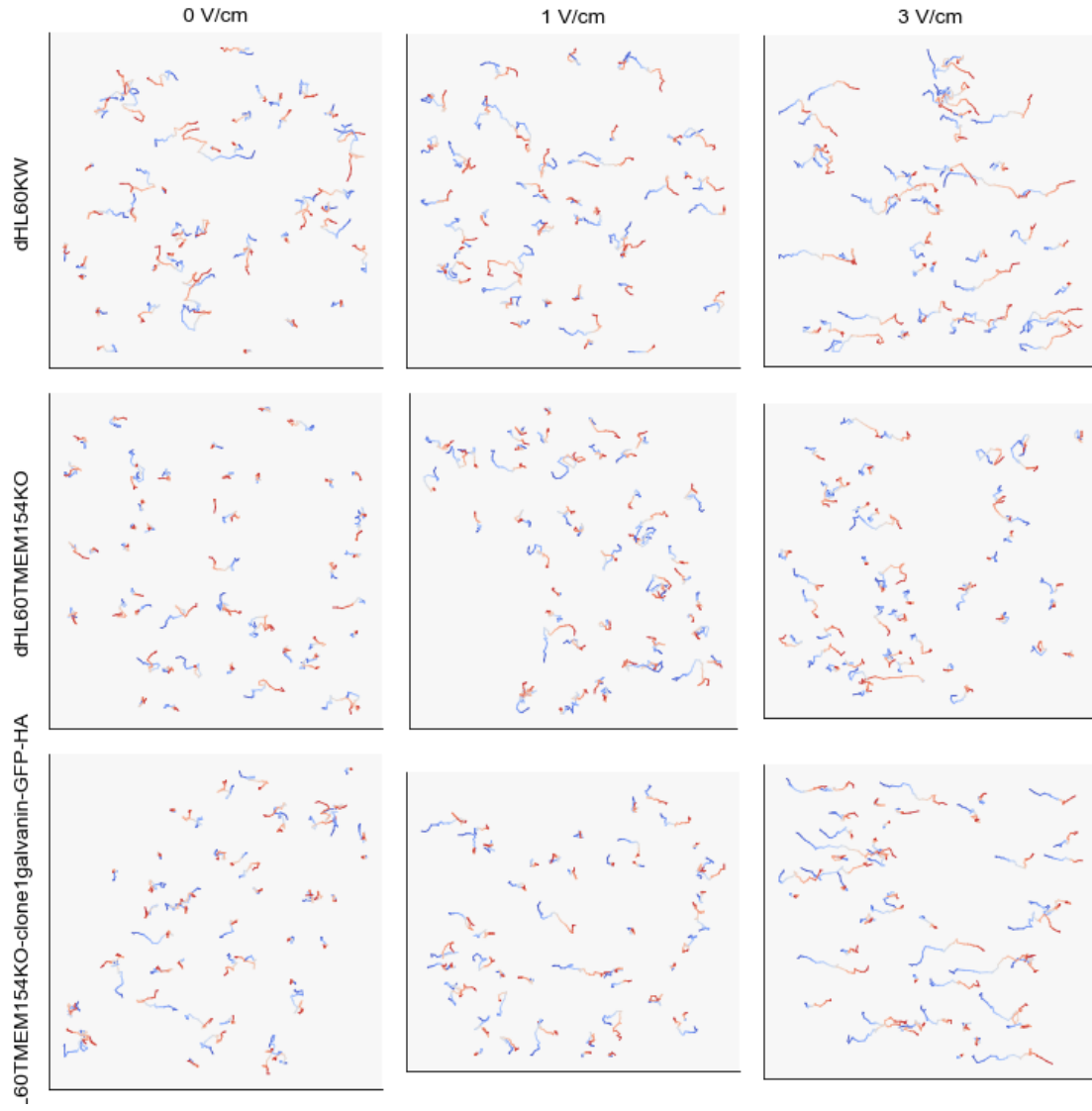
June 8, 2025

```
[11]: array(['dHL60KW', 'dHL60TMEM154KO',  
          'dHL60TMEM154KO-clone1galvanin-GFP-HA'], dtype=object)
```

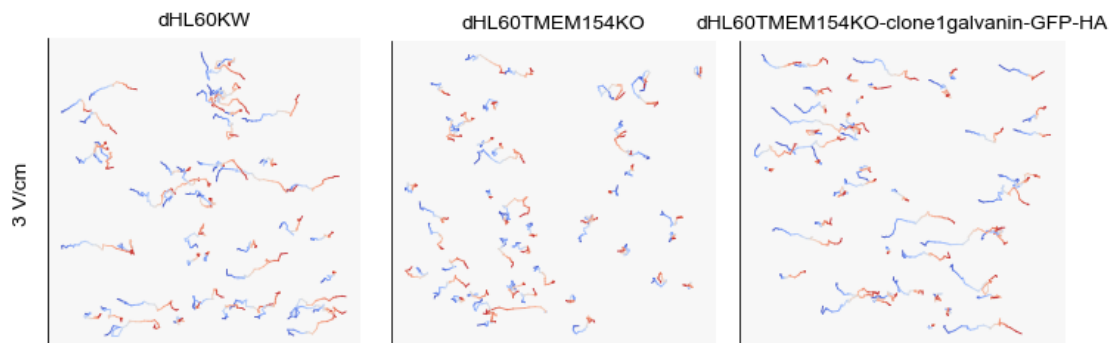
1 Plotting of the HL-60 Galvanin knockout and rescue data.(i.e. Figure 2 data)

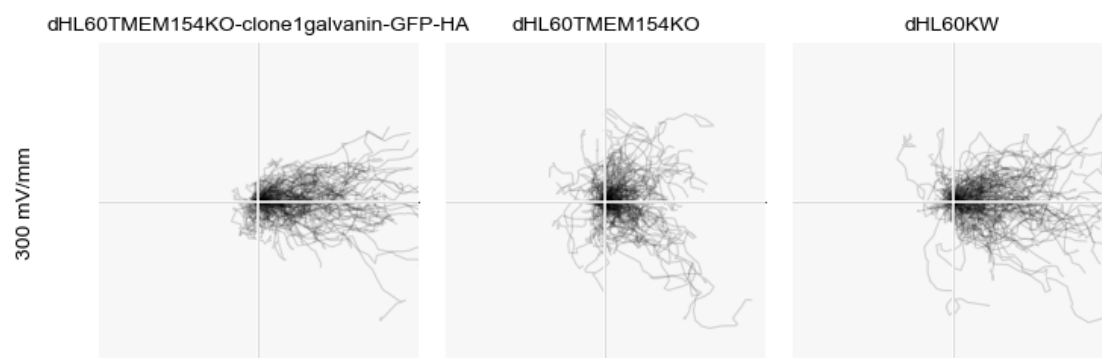
1.1 Tracks



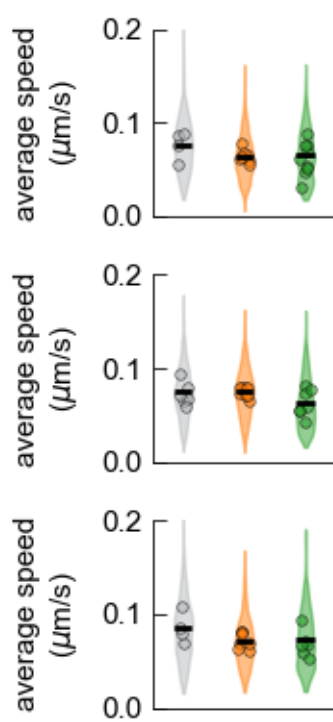


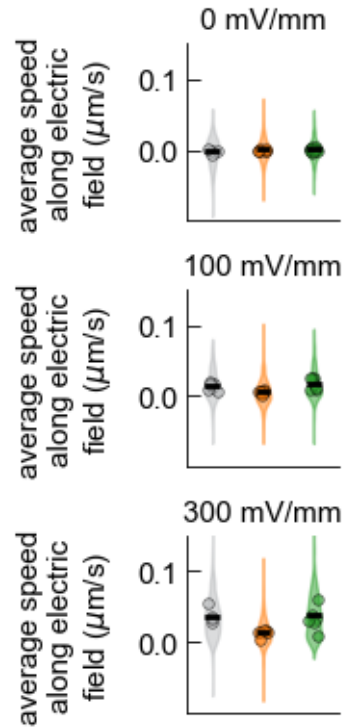
1.2 Here are only plots of 300 mV/mm condition only - probably more useful for main text figures.





2 Speed related



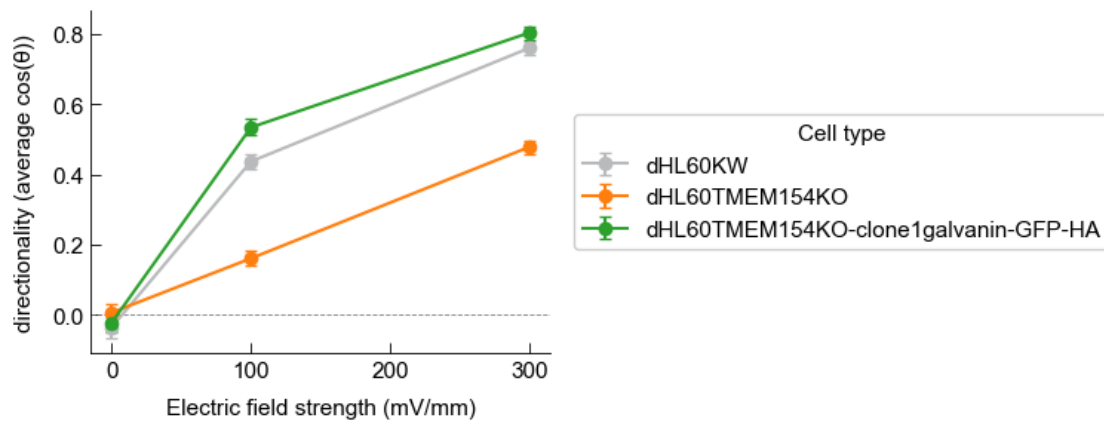


3 Directionality (average cosine theta)

/var/folders/hs/8f67z23n6g9g9g4y03yhrn_c0000gn/T/ipykernel_60724/1976529988.py:5

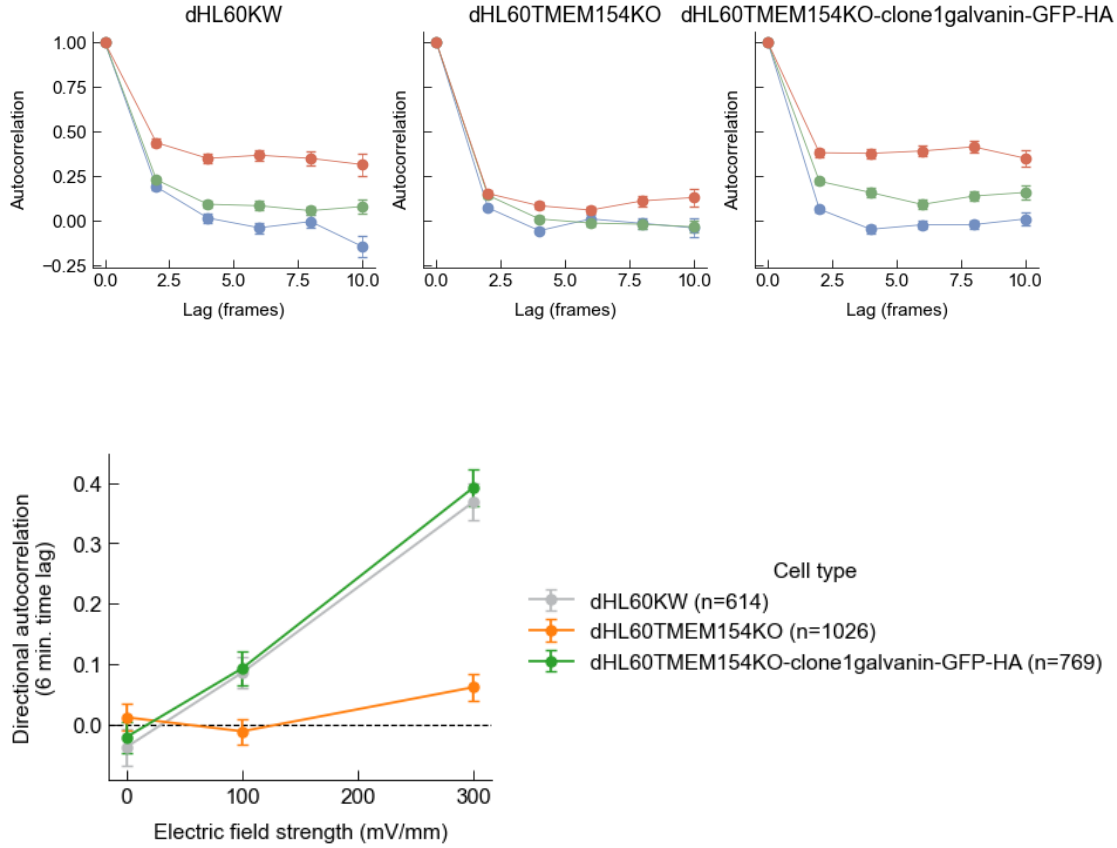
2: UserWarning: Tight layout not applied. The left and right margins cannot be made large enough to accommodate all Axes decorations.

```
plt.tight_layout(rect=[0, 0, 0.8, 1])
```



4 Autocorrelation Analysis

This uses cosine theta between trajectory vector and electric field (the regular approach we've been taking).



4.1 Autocorrelation, but relative to 0 mV/mm condition.

Here we plot relative to the 0 mV/mm condition to try to account for residual persistence in the absence of an electric field. (i.e. subtraction of autocorrelation value from the 0 mV/mm condition).

/var/folders/hs/8f67z23n6g9g9g4y03yhrn_c0000gn/T/ipykernel_60724/195124339.py:35

: UserWarning: Tight layout not applied. The left and right margins cannot be made large enough to accommodate all Axes decorations.

```
plt.tight_layout()
```

